Natural Computing

LMU Munich
summer term 2025

Thomas Gabor

Stirring Soup

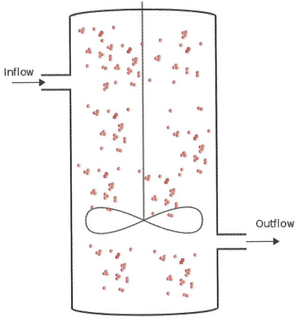


Figure 7.3 Well-stirred flow reactor with dilution — inflow and outflow of material.

Stochasticity

No-Ops, Creation, and Destruction

Example 4. A soup of functions writing into each other...?
Let $\mathcal{X} = \mathcal{X} \to \mathcal{X}$, i.e., the space of all functions that can take functions and return new functions.
Let $\mathcal{R} = \{x_1 + x_2 \rightharpoonup x_1 + x_3 \mid x_1, x_2, x_3 \in \mathcal{X} \wedge x_3 = x_1(x_2)\}$, i.e., when two functions x_1, x_2 meet, x_2 is overwritten with the result of applying x_1 to x_2.
Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.
Let $X_0 \sim \wp^*(\mathcal{X})$.

Example:
λ-calculus

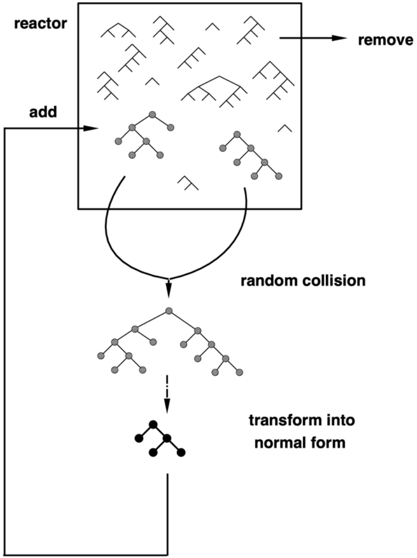


Figure 1: λ-calculus flow reactor. Two expressions A and B are chosen at random and a new object, $(A)B$, is constructed by "application" (see appendix A.2). Putting $(A)B$ into its normal form by β-reduction (see appendix A.2), effectively decides which object species (i.e., "stable molecular formula") the new object is an instance of.

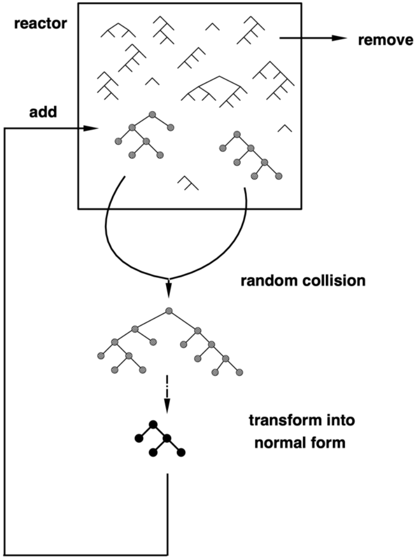
Fontana, Buss. The Barrier of Objects: From Dynamical Systems to Bounded Organizations. 1996.

On λ-calculus...

Example: $$(\lambda x. f\ x)\ y \quad \rightarrowtail \quad f\ y$$

Example in Python: `(lambda x: f(x))(y)` $\rightarrowtail$ `f(y)`

Example:
λ-calculus



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Experiment Time??

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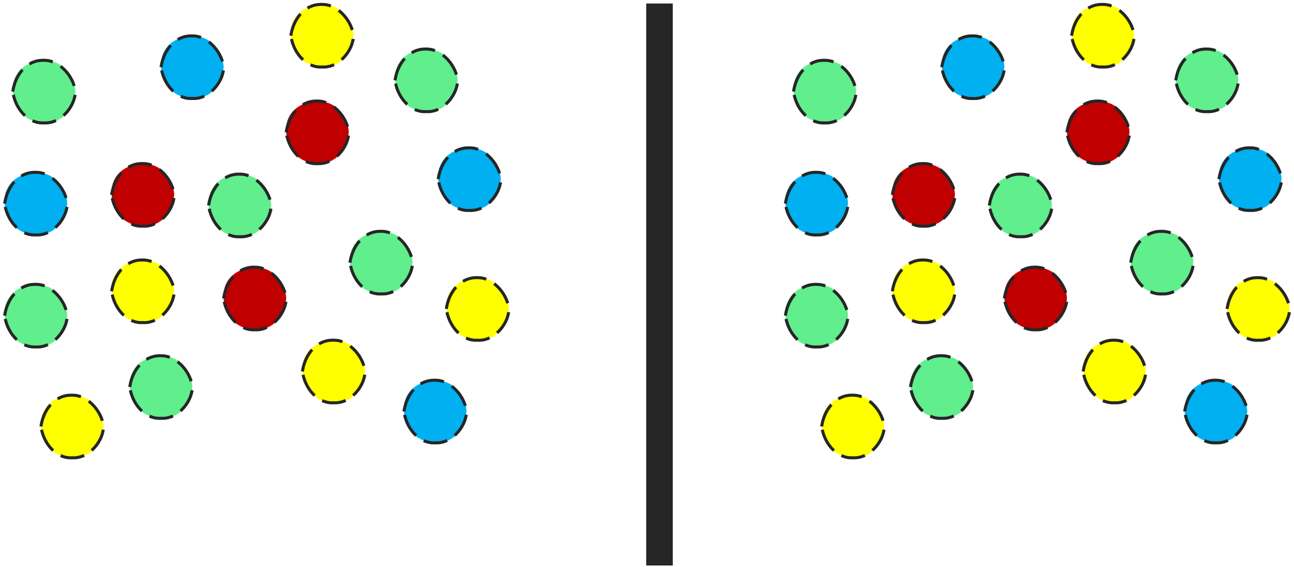
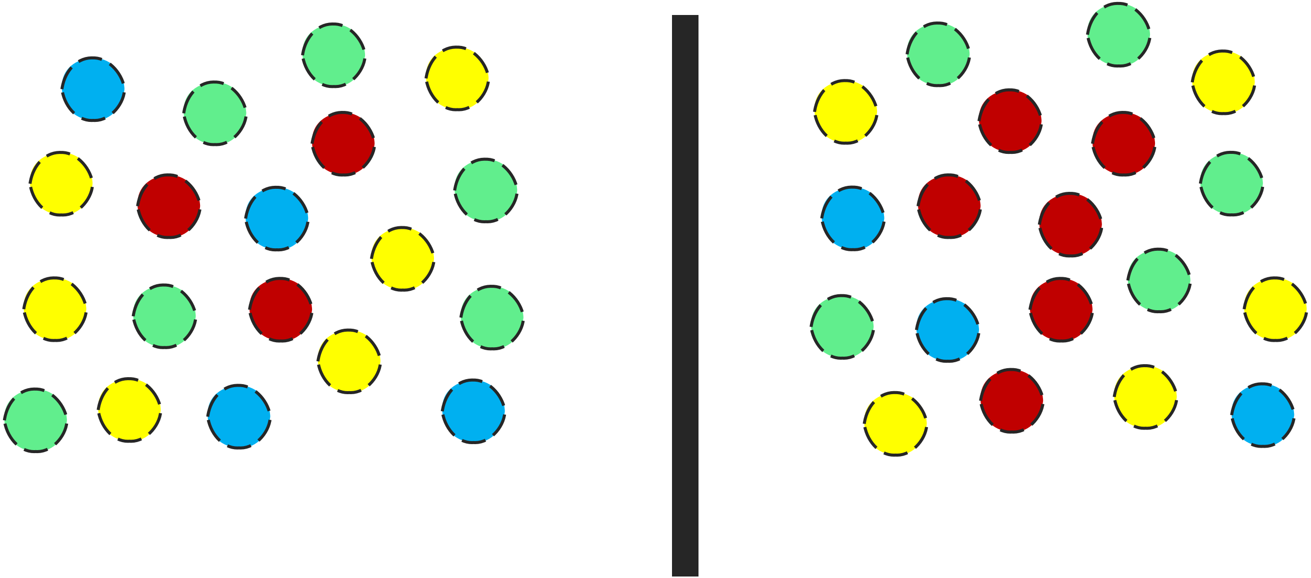
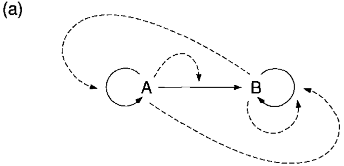
Mere Stability vs. Self-Replication



Mere Stability vs. Self-Replication



Example:
λ-calculus



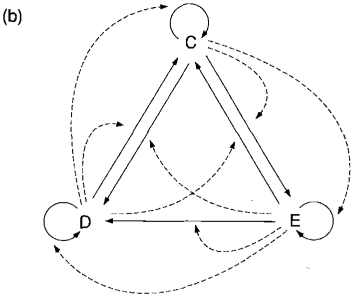


Figure 1. Level 0 ecology. A solid arrow denotes a transformation from argument object to value object. A dotted arrow connects the operator object with the transformation performed by it. Thus, $E \circ C = D$ is shown as a solid arrow from C to D pointed at by a dotted arrow originating in E.
$A \equiv \lambda x_1 . (x_1)\lambda x_2 . \lambda x_3 . (x_3)\lambda x_4 . \lambda x_5 . (x_5)x_4$, $B \equiv \lambda x_1 . (x_1)\lambda x_2 . \lambda x_3 . (x_3)x_2$,
$C \equiv \lambda x_1 . \lambda x_2 . (x_1)\lambda x_3 . x_1$, $D \equiv \lambda x_1 . \lambda x_2 . \lambda x_3 . (x_2)\lambda x_4 . x_2$,
$E \equiv \lambda x_1 . \lambda x_2 . \lambda x_3 . \lambda x_4 . (x_3)\lambda x_5 . x_3$.

Fontana, Buss. The Barrier of Objects: From Dynamical Systems to Bounded Organizations. 1996.

Properties of Self-Replicators

longevity

fecundity

Copy-fidelity

Dawkins. The Selfish Gene. 1976.

Levels of Organization

Level 0: reproduction and ecology

- cycle of replicators (easily collapses to only one replicator)

Level 1: self-maintenance and organization

- emerges when mere copy actions are forbidden
- structure and interaction between particles can be described with specific rules
- part of the organization can generate all its elements anew

Level 2: organization of organizations

- create L1 organizations independently and then put them into the same soup
- organizations maintain themselves separately but interact via "glue" particles

Fontana, Buss. "The Arrival of the Fittest": Toward a Theory of Biological Organization. 1994.

Emergence of
Implicit Target Functions

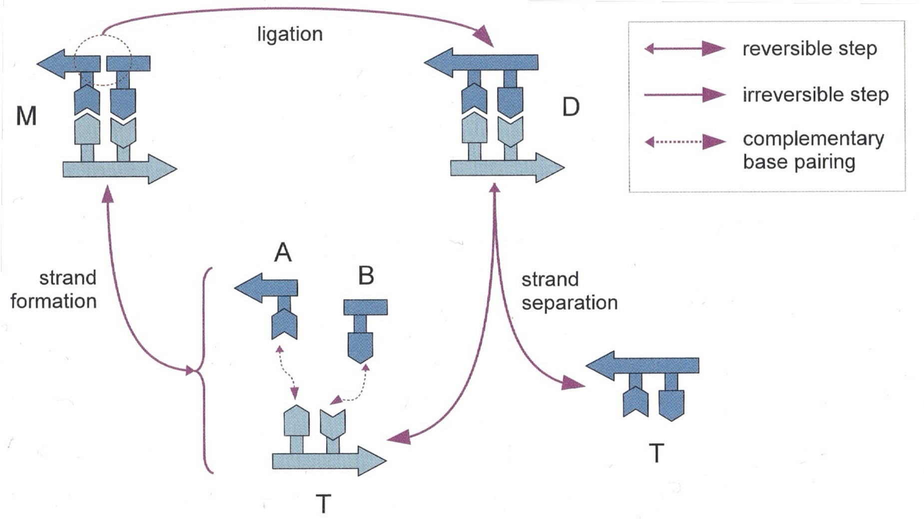
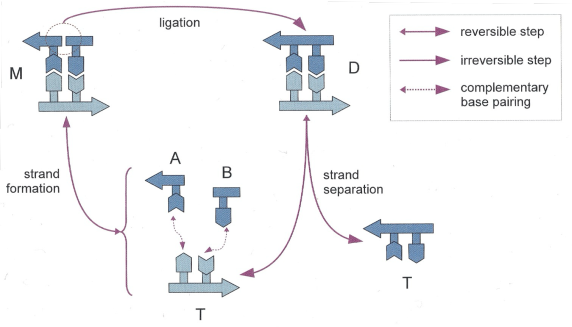


Figure 6.6 A model of the minimal self-replicator demonstrated by Von Kiedrowski [890]: The two T molecules are complementary templates (hexamers in the experiments). Each of them base-pairs with units A and B (trimers in the experiments), to form an intermediate molecule M which is almost a double strand: the still missing link between A and B is then catalyzed by T itself, forming the double strand D. D then splits into two single strands T and the cycle repeats. Adapted from [890].



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Example 5. A DNA soup.

Let $\mathcal{X} = \{A, C, G, T\}^{2\times 2} \cup \{A, C, G, T\}^2 \cup \{A, C, G, T\}^1$.

Let $\mathcal{R} = \{R_{1,\_,\_}, R_{2,\_,\_}, R_{3,\_,\_}, R_{4,\_}\}$ with

$$
\begin{aligned}
R_{1,X,Y} &= (X \quad Y) + (-X) + (-Y) && \rightharpoonup^{0.5} && \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} \\
R_{2,X,Y} &= \begin{pmatrix} X & Y \\ -X & -Y \end{pmatrix} && \rightharpoonup && (X \quad Y) + (-Y \quad -X) \\
R_{3,X,Y} &= (X) + (Y) && \rightleftharpoons^{0.01}_{0.01} && (X \quad Y) \\
R_{4,X} &= \emptyset && \rightharpoonup^{0.1} && (X)
\end{aligned}
$$

for all $X, Y \in \{A, C, G, T\}$.

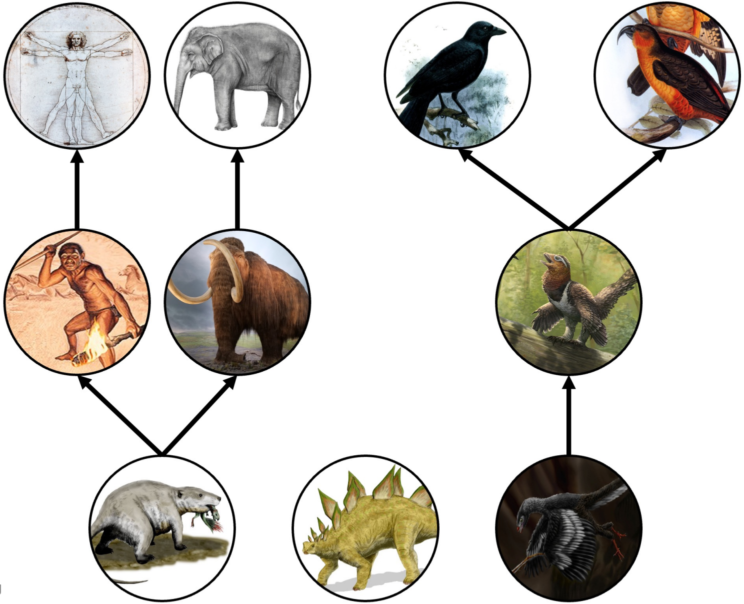
Let $A(X) \sim \mathcal{A}(X, \mathcal{R})$.

Let $X_0 = \emptyset$.

Natural Evolution

"The history of the universe can be thought of as a sequence of information processing revolutions, each of which builds on the technology of the previous ones."

Seth Lloyd. Programming the Universe. Vintage Books, 2006.



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upload.wikimedia.org/wikipedia/commons/6/69/Balaur_bondoc_as_avialan.jpg
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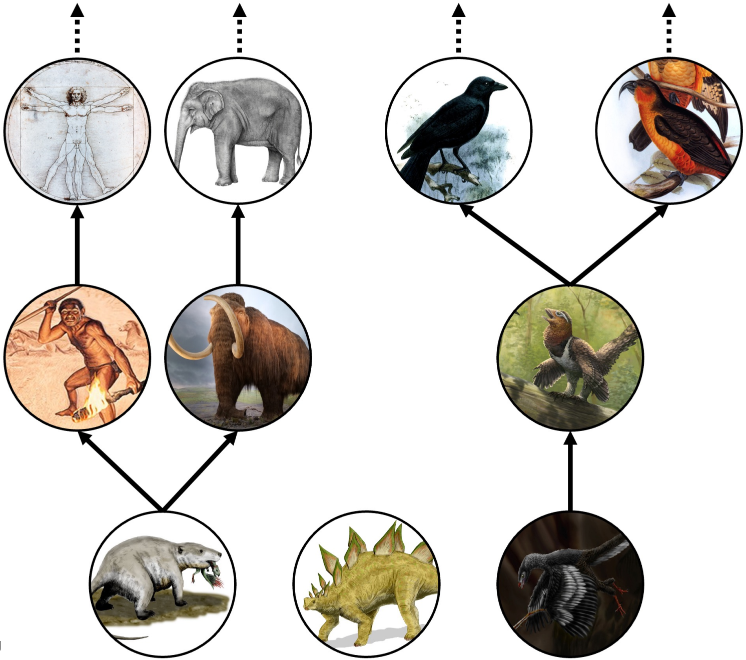
Why does it keep going?



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upload.wikimedia.org/wikipedia/commons/6/69/Balaur_bondoc_as_avialan.jpg
indianapublicmedia.org/amomentofscience/crow-thievery-2.php

The Subject of Evolution

Dawkins. The Selfish Gene. Oxford University Press, 1976.

The Subject of Evolution

The Gene

in the evolutionary biology sense

Dawkins. The Selfish Gene. Oxford University Press, 1976.

Other
(Non-Biological)
Instances of Evolution?

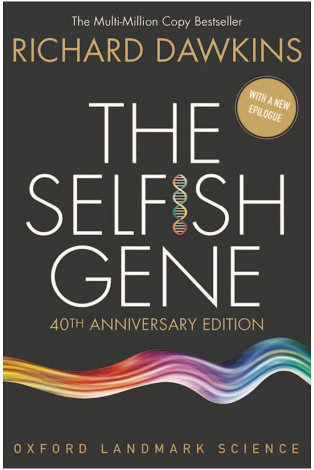
Dawkins. The Selfish Gene. Oxford University Press, 1976.

Dawkins.

The Selfish Gene. (40th Anniversary Edition)

Oxford University Press, 2016.

ebookcentral.proquest.com/lib/ub-lmu/detail.action?docID=4545419



Evolutionary Computing

Definition 11 (population-based optimization)**.** Let $\mathcal{X}$ be a state space. Let $\mathcal{T}$ be a target space with total order $\leq$. Let $\tau : \mathcal{X} \to \mathcal{T}$ be a target function. A tuple $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_t \rangle_{0 \leq t \leq g})$ is a population-based optimization process iff $X_t \in \wp^*(\mathcal{X})$ for all t and $E : \langle \wp^*(\mathcal{X}) \rangle \times (\mathcal{X} \to \mathcal{T}) \to \wp^*(\mathcal{X})$ is a possibly randomized, non-deterministic, or further parametrized function so that the population-based optimization run is produced by calling E repeatedly, i.e., $X_{t+1} = E(\langle X_u \rangle_{0 \leq u \leq t}, \tau)$ where X_0 is given externally or chosen randomly.

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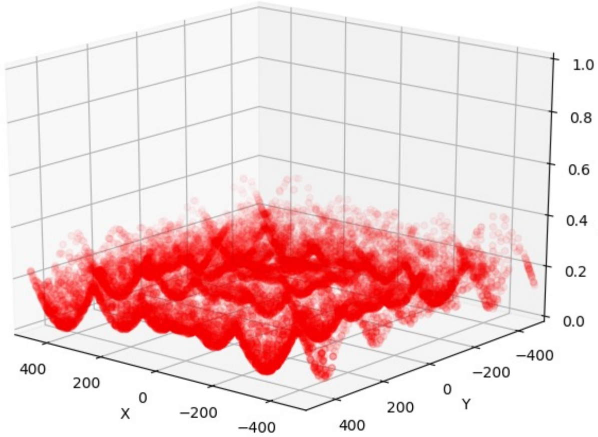
Definition 12 (population-based optimization (alternate))**.** An optimization process $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_t \rangle_{0 \leq t \leq g})$ is called population-based iff $\mathcal{X}$ has the form $\mathcal{X} = \wp^*(\mathcal{Y})$ for some other state space $\mathcal{Y}$.

Algorithm 6 (basic evolutionary algorithm)**.** Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \tau, E, \langle X_u \rangle_{0 \leq u \leq t})$ be a population-based optimization process. The process $\mathcal{E}$ continues via an evolutionary algorithm if E has the form

$$E(\langle X_u \rangle_{0 \leq u \leq t}, \tau) = X_{t+1} = \mathit{selection}\big(X_t \uplus \mathit{variation}(X_t)\big)$$

where *selection* and *variation* are possibly randomized or non-deterministic functions so that for any $X \in \wp^*(\mathcal{X})$ it holds that $|\mathit{selection}(X)| \leq |X|$ and $\big|\mathit{selection}\big(X \uplus \mathit{variation}(X)\big)\big| = |X|$.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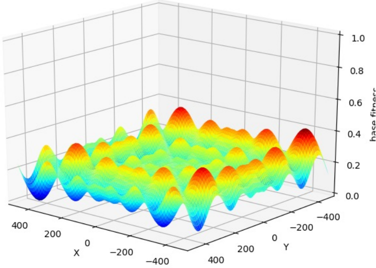
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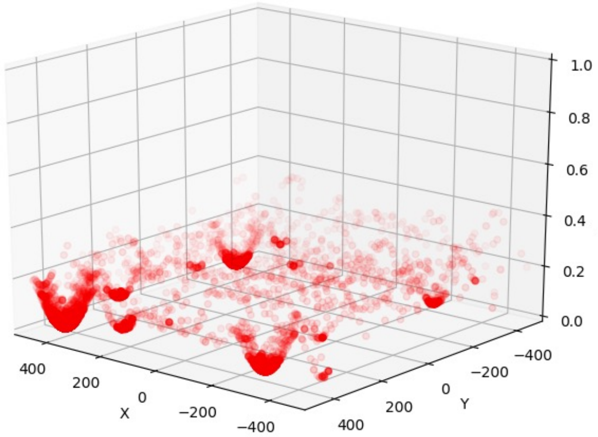
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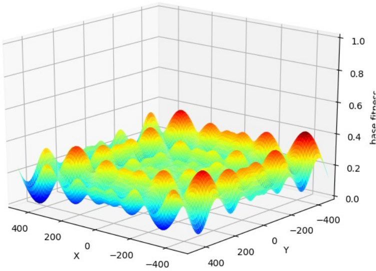
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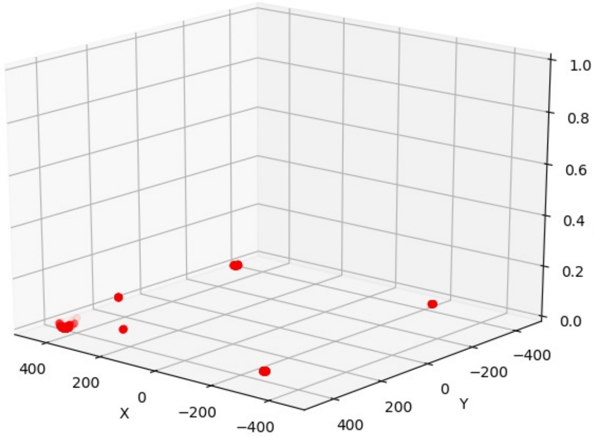
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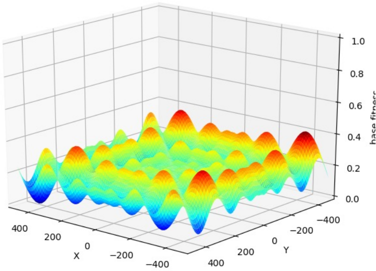
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Variation

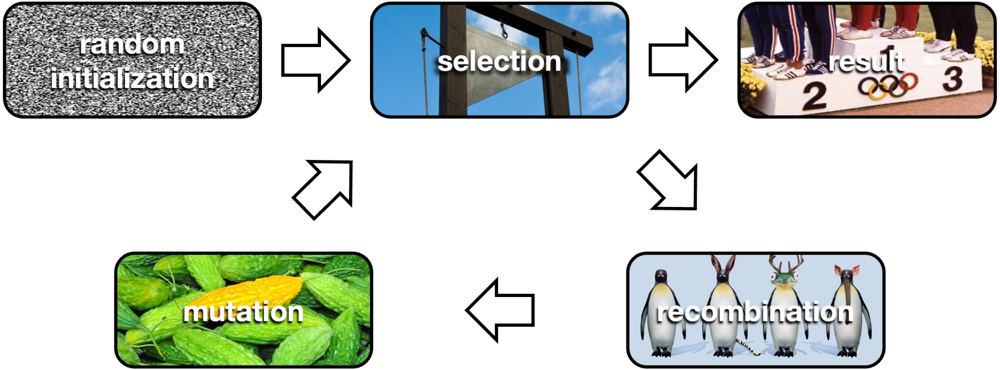
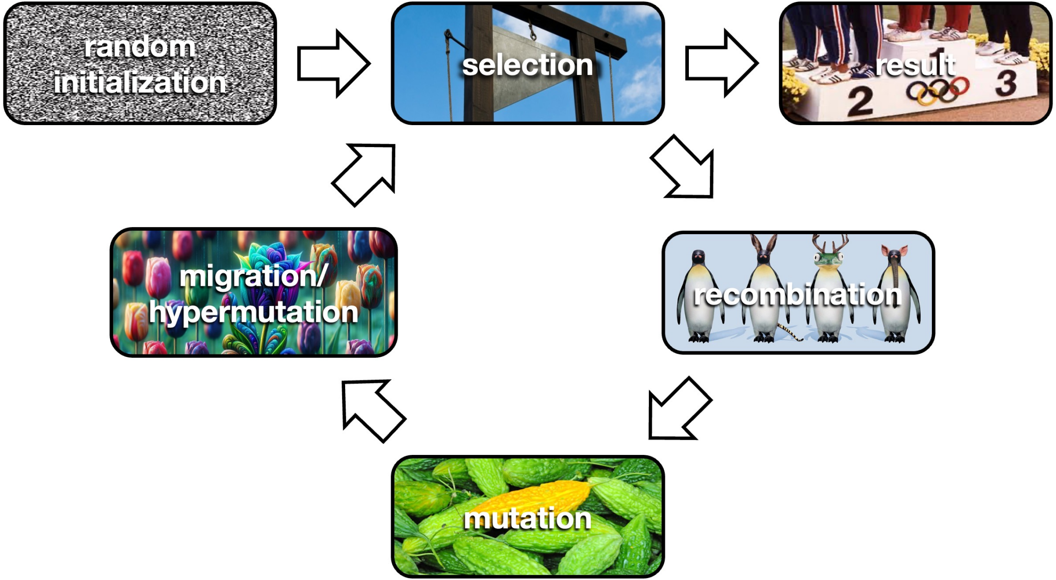


image sources:
www.bostonmagazine.com/news/2015/07/30/boston-2024-winners-losers
en.wikipedia.org/wiki/Mutation#/media/File:Darwin_Hybrid_Tulip_Mutation_2014-05-01.jpg
www.heise.de/ct/artikel/Die-Woche-Microsoft-und-Linux-1283059.html
https://phys.org/news/2019-10-guillotine-cruel-poisoning.html

Recombination

Mutation

Migration/Hypermutation



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chat.openai.com

Natural vs. Artificial Selection

Target Function vs. Fitness Function

Points of Selection

Types of Selection

Cut-Off Selection

Types of Selection

Tournament Selection

Types of Selection

Roulette Wheel Selection

Algorithm 7 (typical evolutionary algorithm)**.** Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \phi, E, \langle X_u \rangle_{0 \leq u \leq t})$ be a population-based optimization process.

- Let $\sigma_N^{survivors}, \sigma_N^{parents} : \wp^*(\mathcal{X}) \to \wp^*(\mathcal{X})$ be (possibly randomized or non-deterministic) selection functions that return $N \in \mathbb{N}$ individuals, i.e., $|\sigma_N(X)| = N$ and $\sigma_N(X) \subseteq X$ for all X. They may possibly depend on $\mathcal{E}$ (most commonly ϕ). Let $\varrho_N^{mutants} : \wp^*(\mathcal{X}) \to \wp^*(\mathcal{X})$ be a special selection function that, given a population X, returns a random subset $X' \subseteq X$ so that $|X'| = N$. Note that $\varrho_{|X|}(X) = X$ for all X.

- Let $mutate : \mathcal{X} \to \mathcal{X}$ be a (possibly randomized or non-deterministic) single-individual mutation function. We write $mutate[\_] : \wp^*(\mathcal{X}) \to \wp^*(\mathcal{X})$ for the per-individual application of $mutate$, i.e., $mutate[X] = \{mutate(x) : x \in X\}$.

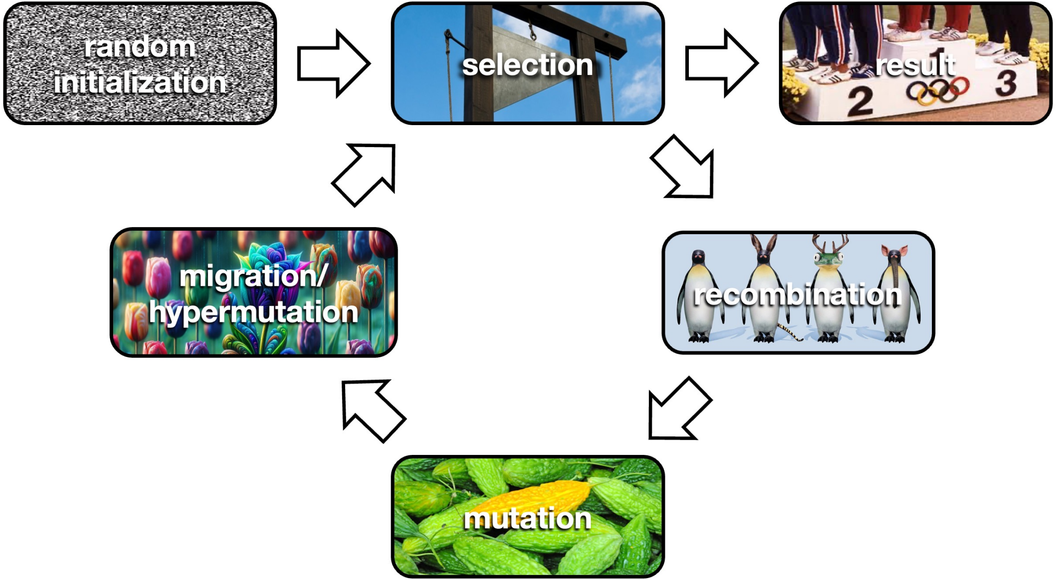
- Let $recombine : \mathcal{X} \times \mathcal{X} \to \mathcal{X}$ be a (possibly randomized or non-deterministic) two-parent recombination function. We write $recombine[\_] : \wp^*(\mathcal{X}) \to \wp^*(\mathcal{X})$ for the random-pairing application of $recombine$, i.e., $recombine[X] = \{recombine(x_1, x_2)\} \uplus recombine[X \setminus \{x_1, x_2\}]$ with $x_1, x_2 \sim X$ and $recombine[\{x\}] = \emptyset$ for all x as well as $recombine[\emptyset] = \emptyset$.

- Let $migrate : \emptyset \to \mathcal{X}$ be a (possibly randomized or non-deterministic) individual creation function. We write $migrate_N : \emptyset \to \wp^*(\mathcal{X})$ with $N \in \mathbb{N}$ for the iterated application of $migrate$, i.e., $migrate_N[] = \{migrate()\} \uplus migrate_{N-1}[]$ with $migrate_0[] = \emptyset$.

The process $\mathcal{E}$ continues via a typical evolutionary algorithm if E has the form

$$\begin{aligned} E(\langle X_u \rangle_{0 \leq u \leq t}, \phi) = X_{t+1} = \sigma_{|X_t|}^{survivors}\big(X_t \\ \uplus\, mutate[\varrho_M^{mutants}(X_t)] \\ \uplus\, recombine[\sigma_{2C}^{parents}(X_t)] \\ \uplus\, migrate_H[]\big) \end{aligned}$$

where $M \in \mathbb{N}$ is the number of mutants produced per generation, $C \in \mathbb{N}$ is the number of children produced per generation, and $H \in \mathbb{N}$ is the number of migrants (sometimes called hypermutants) generated per generation. Especially M is often expressed as a rate for random choice within the population, i.e., $M = \lceil m \cdot |X_t| \rceil$ where $m \in \mathbb{R}$ is called mutation rate.



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Exploration/Exploitation for EAs

Diversity

Algorithm 8 (diversity-aware evolutionary algorithm)**.** Let $\mathcal{E} = (\mathcal{X}, \mathcal{T}, \phi, E, \langle X_u \rangle_{0 \leq u \leq t})$ be the process of an evolutionary algorithm (cf. Alg. 7). $\mathcal{E}$ is called a diversity-aware evolutionary algorithm if the used fitness function $\phi : \mathcal{X} \to \mathcal{T}$ has the form

$$\phi(x) = (1 - \zeta) \cdot \tau(x) + \zeta \cdot \delta(x)$$

where $\tau : \mathcal{X} \to \mathcal{T}$ is the underlying target function of the given optimization problem, $\delta : \mathcal{X} \to \mathcal{T}$ is a (possibly randomized, non-deterministic, or further parametrized) function that estimates the diversity that a given individual contributes to the population, and $\zeta \in [0; 1] \subset \mathbb{R}$ is called a diversity weight.
Note that according to Thm. 4 and Gabor et al. [2] the diversity estimate

$$\delta(x) = \frac{1}{|X_t|} \cdot \sum_{x' \in X_t} |x' - x|$$

is generally suitable for $\mathcal{X} = \mathbb{R}^n$ or $\mathcal{X} = \mathbb{Z}^n$ for some n.

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Theorem 4 (Wineberg-Oppacher diversity [4])**.** All measures of the diversity of an individual x in a population X are related to the pairwise distance between individuals in X as measured by a suitable norm (Euclidean for real-valued representations, Hamming for symbolic representations, e.g.).

The Ideal Fitness Function?